

The empirical recurrence for OEIS A237136: recovering a conjecture that is not written down

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Abstract

OEIS A237136 records “Empirical recurrence of order 44”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 216$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 44, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 44 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237136 is “Number of $(n+1) \times (2+1)$ 0..3 arrays with the maximum plus the upper median of every 2×2 subblock equal.”. It has offset 1 and begins

1500, 22960, 333446, 5961802, 107652788, 2051159992, 39515565068, 773779308602, ...

The entry states:

Empirical recurrence of order 44 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:52:17, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 216$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 216$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 432$ exact terms of the model returns order 44 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{44} c_i a(n-i),$$

c_1	46	c_{12}	174699644591	c_{23}	−153010566432611284	c_{34}	96771124641269418624
c_2	−405	c_{13}	−16542084024522	c_{24}	−5685917526198766996	c_{35}	−271194975230444439552
c_3	−10536	c_{14}	24177332907723	c_{25}	3390281858749645616	c_{36}	−29774082584430360576
c_4	198841	c_{15}	299478817623450	c_{26}	20869732076084124552	c_{37}	195610669172800229376
c_5	237486	c_{16}	−820151283416609	c_{27}	−19616966929262012240	c_{38}	−16527290212399841280
c_6	−25811755	c_{17}	−3442054482156536	c_{28}	−54719769864022083320	c_{39}	−87291392101839175680
c_7	101825398	c_{18}	14000351851106089	c_{29}	67709845549376736288	c_{40}	19405814564841062400
c_8	1527804956	c_{19}	24200865414788090	c_{30}	101058233914776507168	c_{41}	21024116656039526400
c_9	−11346994706	c_{20}	−150914215569177206	c_{31}	−156480034027229989824	c_{42}	−6689370148503552000
c_{10}	−42294875090	c_{21}	−82918225679545688	c_{32}	−126411449278545095520	c_{43}	−1950814915264512000
c_{11}	569546077698	c_{22}	1105105833909085814	c_{33}	249014156453218782720	c_{44}	735578286981120000

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 44, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $44 + 4$ terms removed returns order 44 as well, so $\deg q_{\min} = 44 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $44 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 44 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A237136.

- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.